

**SHETH L.U.J AND SIR M.V. COLLEGE**  
**SUBJECT: DATA VISUALIZATION WITH POWER BI AND TABLEAU**

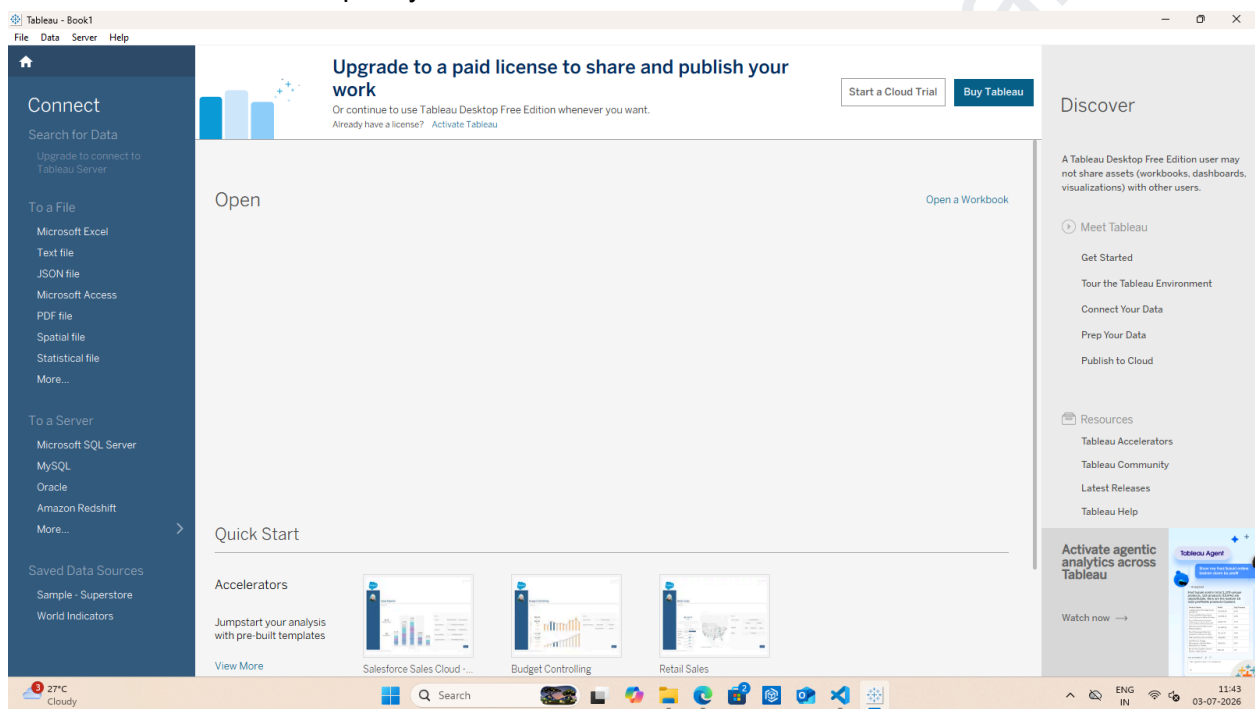
**AIM: Practical 1: Getting Started with Tableau and Building Basic Visualizations**

- a) Connect to a CSV dataset and explore the Tableau interface
- b) Identify and differentiate Measures and Dimensions
- c) Convert fields between discrete and continuous and observe changes
- d) Create and customize a basic bar chart
- e) Build a line chart for time-series analysis
- f) Create a geographic map visualization
- g) Combine multiple charts into a simple interactive dashboard

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**a) Connect to a CSV dataset and explore the Tableau interface:**

1. Launch Tableau Desktop on your machine .



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2. Select your CSV file from your Mac Finder and click Open.

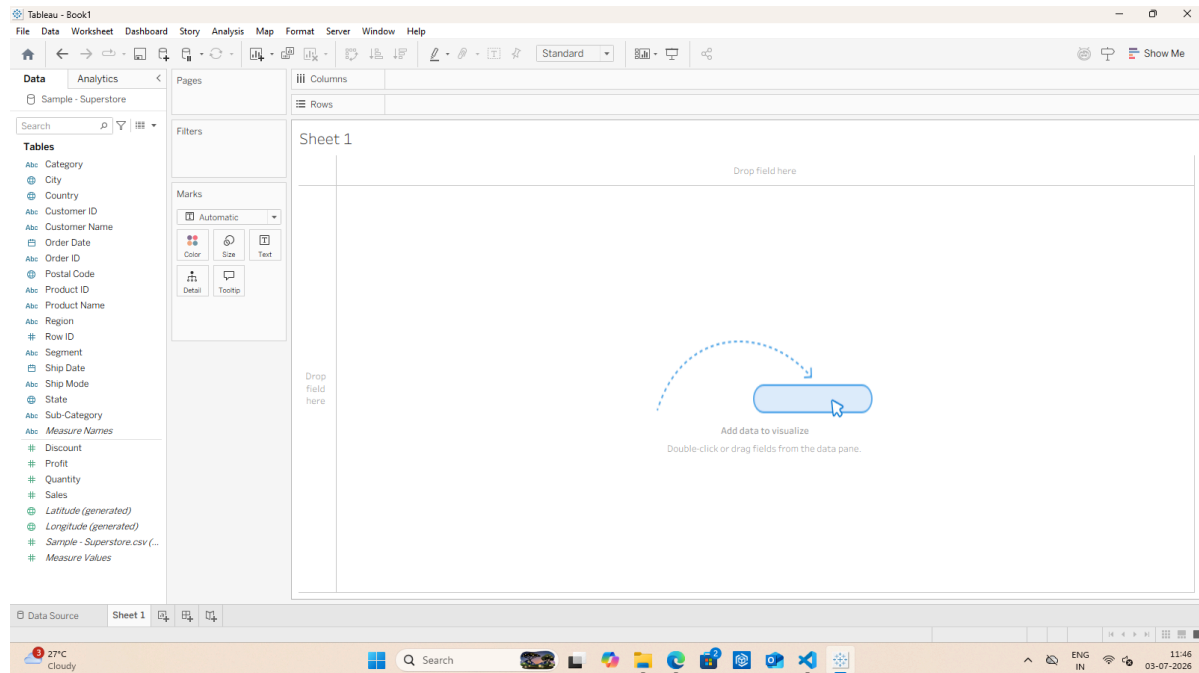
The screenshot shows the Tableau Desktop interface with the 'Sample - Superstore' data source loaded. The sidebar on the left shows the 'Connections' and 'Files' sections. The main area displays a preview of the CSV data with columns like Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, and Customer Name. The bottom status bar shows 'Data Source' and 'Sheet 1'.

| Row ID | Order ID       | Order Date | Ship Date  | Ship Mode      | Customer ID | Customer Name |
|--------|----------------|------------|------------|----------------|-------------|---------------|
| 1      | CA-2016-152156 | 08-11-2016 | 11-11-2016 | Second Class   | CG-12520    | Claire Gul    |
| 2      | CA-2016-152156 | 08-11-2016 | 11-11-2016 | Second Class   | CG-12520    | Claire Gul    |
| 3      | CA-2016-138688 | 12-06-2016 | 16-06-2016 | Second Class   | DV-13045    | Darrin Var    |
| 4      | US-2015-108966 | 11-10-2015 | 18-10-2015 | Standard Class | SO-20335    | Sean O'Di     |
| 5      | US-2015-108966 | 11-10-2015 | 18-10-2015 | Standard Class | SO-20335    | Sean O'Di     |
| 6      | CA-2014-115812 | 09-06-2014 | 14-06-2014 | Standard Class | BH-11710    | Brosina H     |
| 7      | CA-2014-115812 | 09-06-2014 | 14-06-2014 | Standard Class | BH-11710    | Brosina H     |
| 8      | CA-2014-115812 | 09-06-2014 | 14-06-2014 | Standard Class | BH-11710    | Brosina H     |
| 9      | CA-2014-115812 | 09-06-2014 | 14-06-2014 | Standard Class | BH-11710    | Brosina H     |
| 10     | CA-2014-115812 | 09-06-2014 | 14-06-2014 | Standard Class | BH-11710    | Brosina H     |
| 11     | CA-2014-115812 | 09-06-2014 | 14-06-2014 | Standard Class | BH-11710    | Brosina H     |
| 12     | CA-2014-115812 | 09-06-2014 | 14-06-2014 | Standard Class | BH-11710    | Brosina H     |
| 13     | CA-2017-114412 | 15-04-2017 | 20-04-2017 | Standard Class | AA-10480    | Andrew A      |
| 14     | CA-2016-161389 | 05-12-2016 | 10-12-2016 | Standard Class | IM-15070    | Irene Mac     |

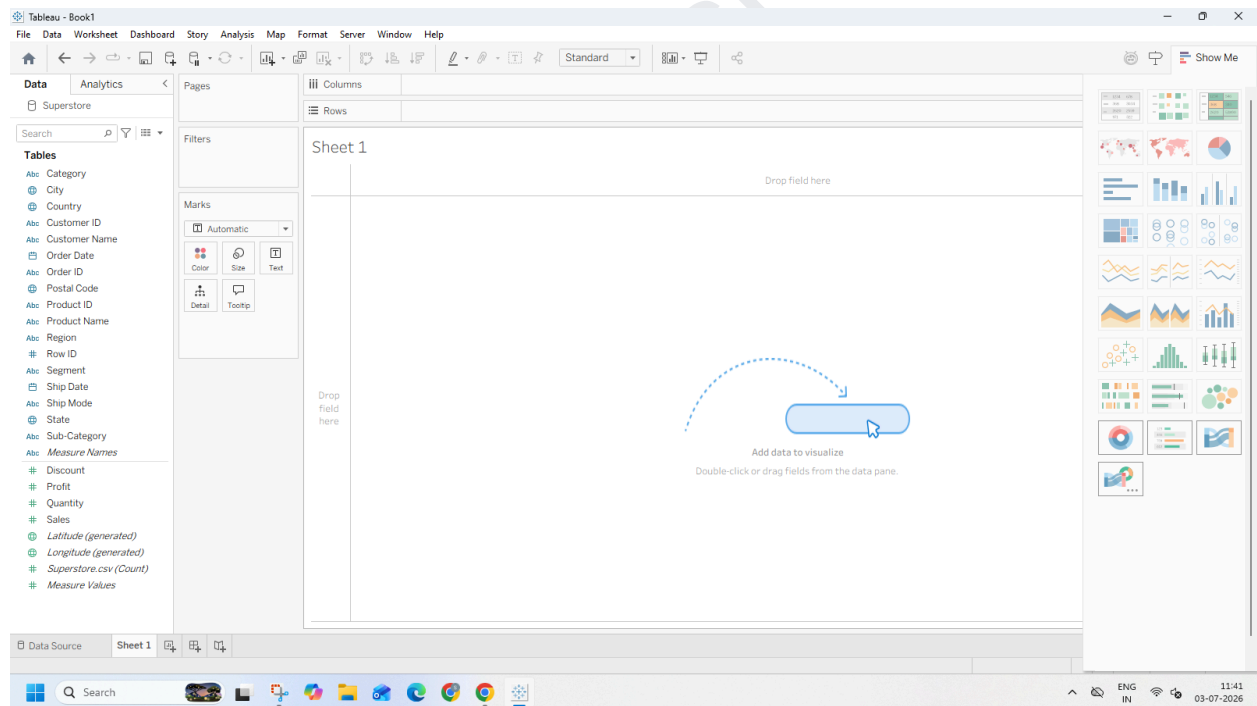
3. You will be taken to the Data Source page. Here, preview your data rows at the bottom to ensure they look correct, then click Sheet 1 at the bottom left to enter your workspace. Tableau will bypass its faulty drivers, create a native workspace called Clipboard, and instantly open up Sheet 1.

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### b) Identify and differentiate Measures and Dimensions:



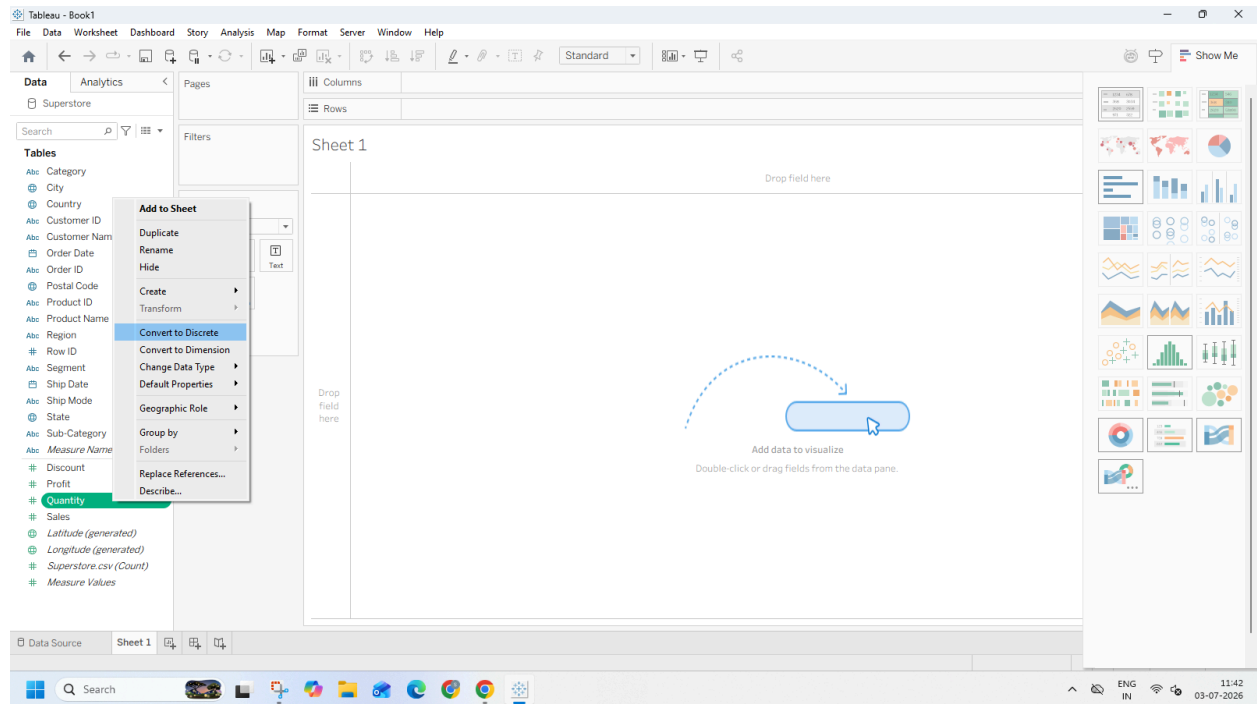
### c) Convert fields between discrete and continuous and observe changes:

1. Converting Quality from Continuous to Discrete.

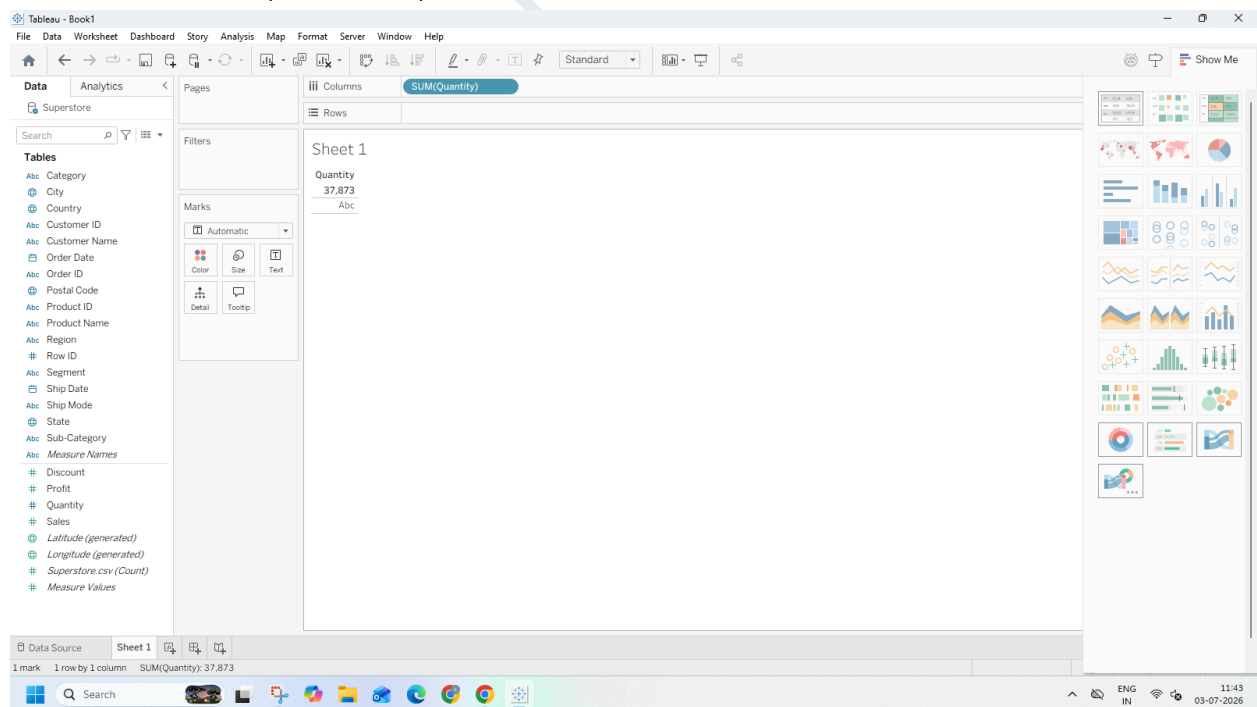
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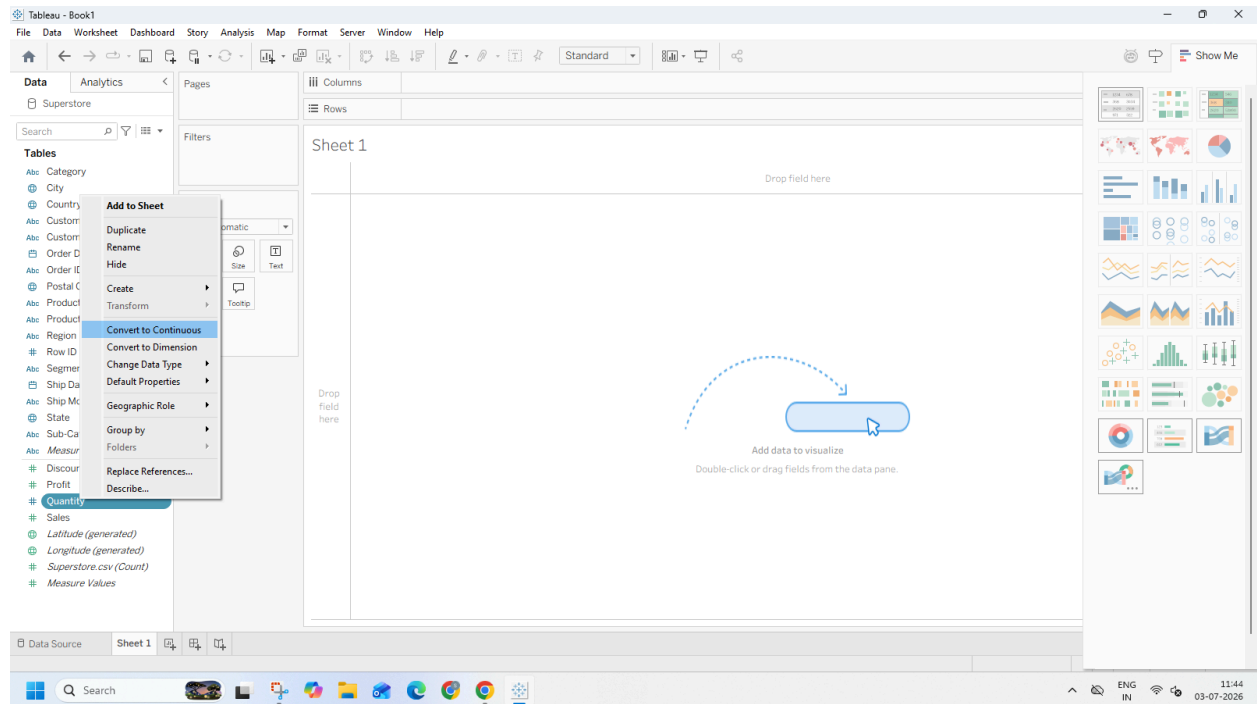
2. Dragging Quantity into Columns shelf. It will draw isolated individual header columns for each exact number (1, 2, 3, etc.).



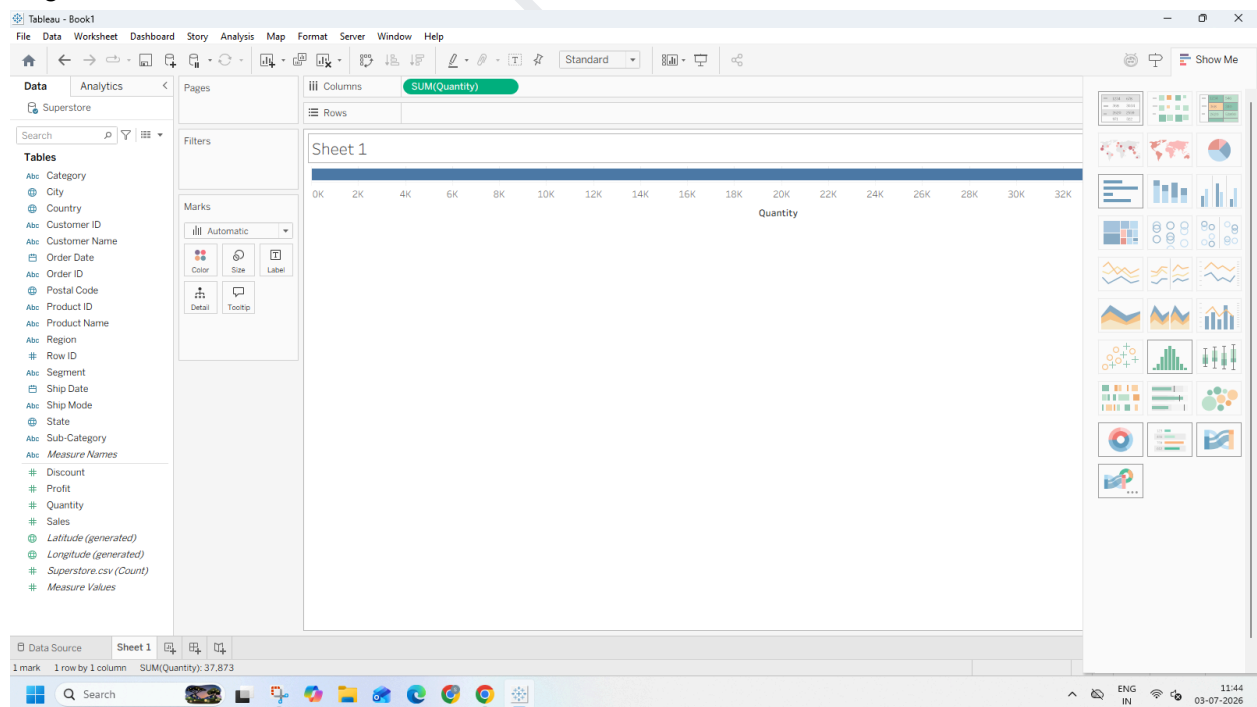
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3. Dragging back it off the shelf. And Converting it from Discrete to Continuous.



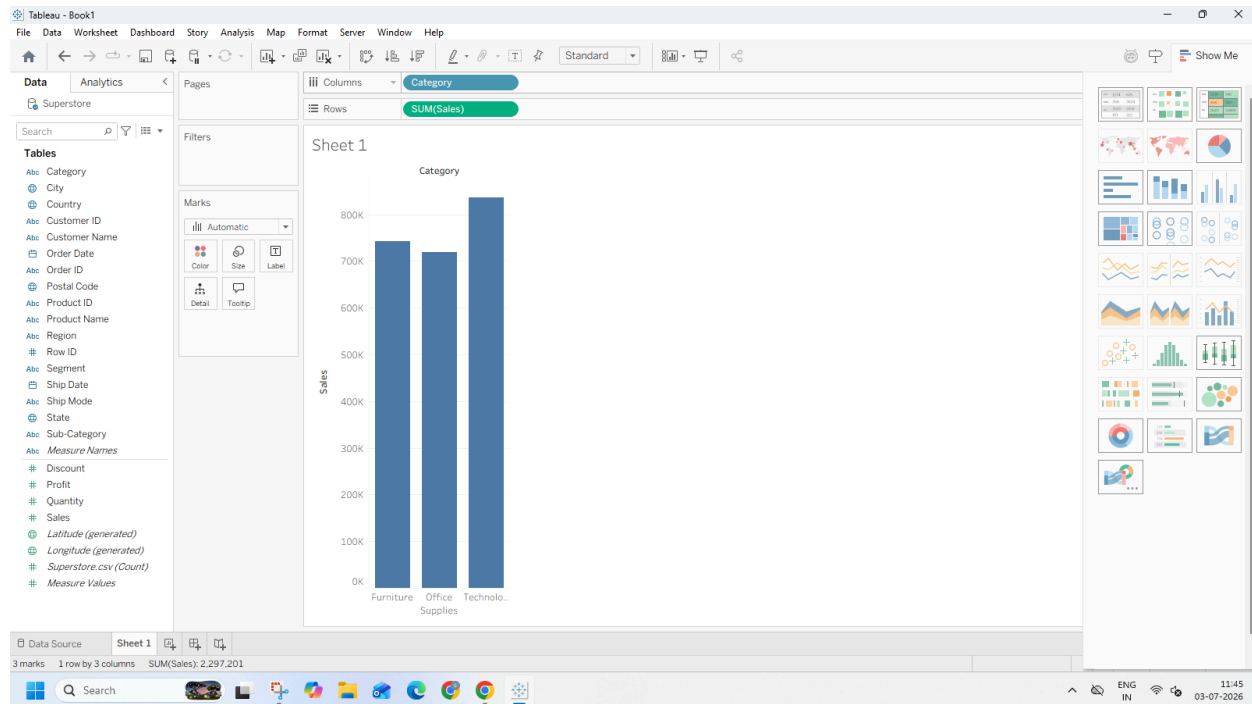
4. Drag it back to Column Shelf, Instead of individual column blocks, Tableau now draws a singular, smooth, continuous numeric axis line.



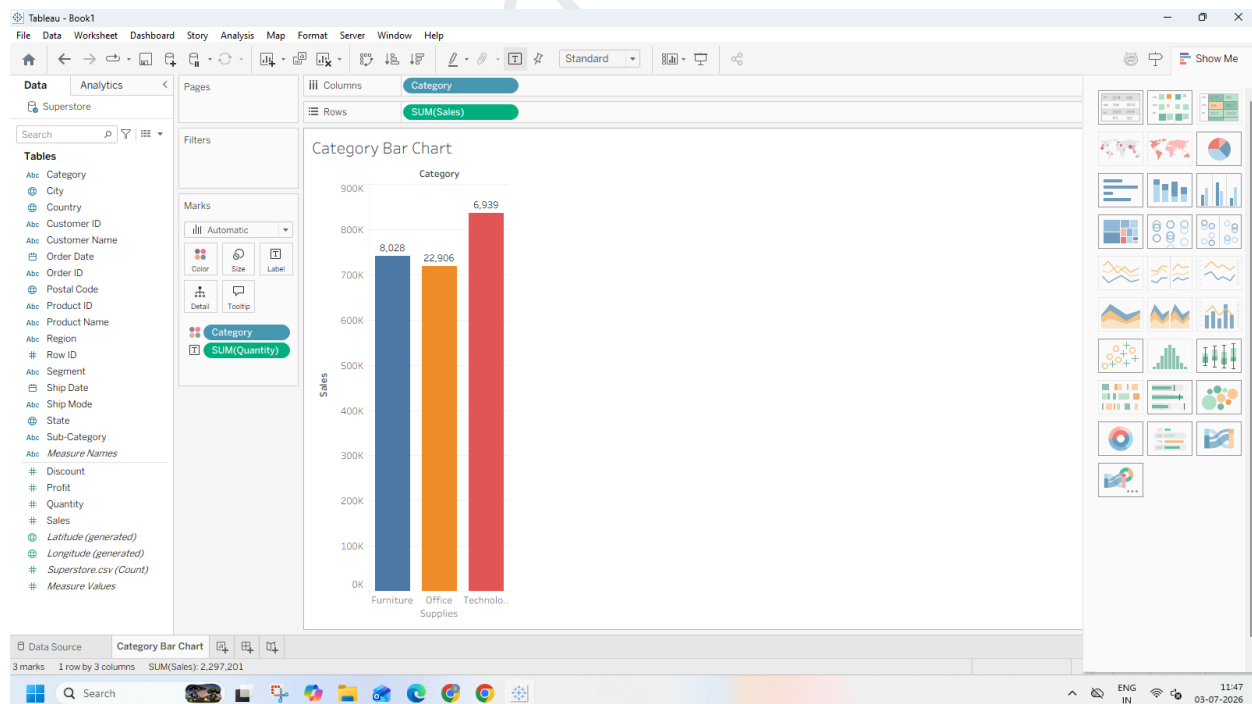
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**d) Create and customize a basic bar chart:**

**1. Dragging Category to Column Shelf and Sales into Row Shelf.**



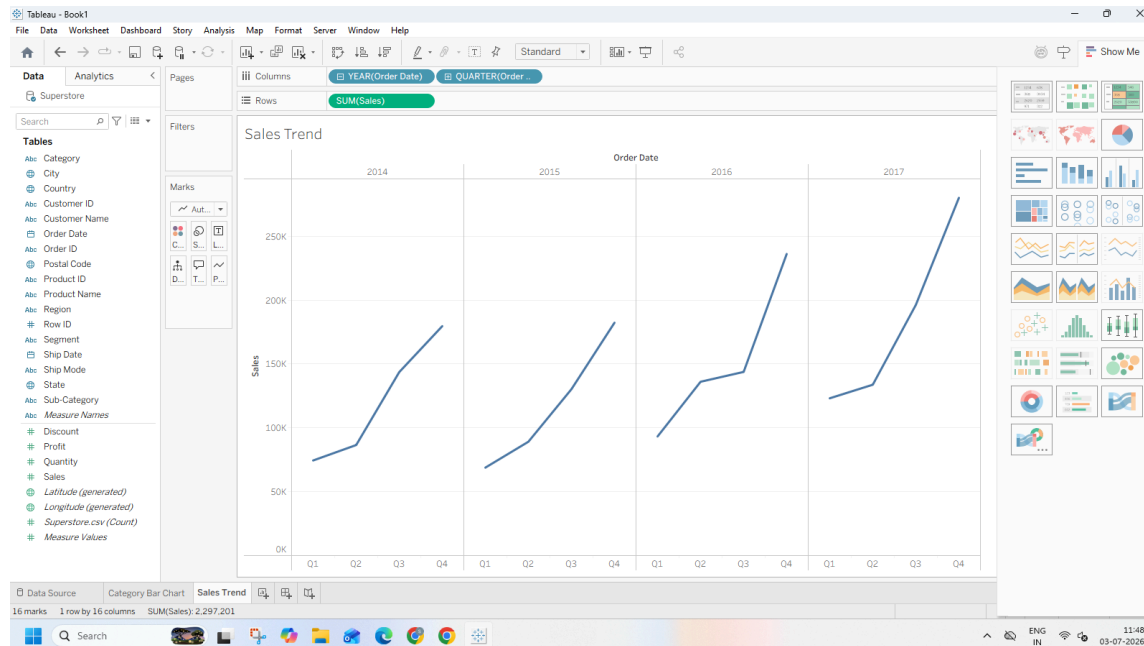
**2. Color customization: Take Category from the Data pane a second time, and drop it directly onto the Color tile in the Marks card. And Double clicking the Sheet to name it 'Category Bar Chart'.**



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**e) Build a line chart for time-series analysis:**

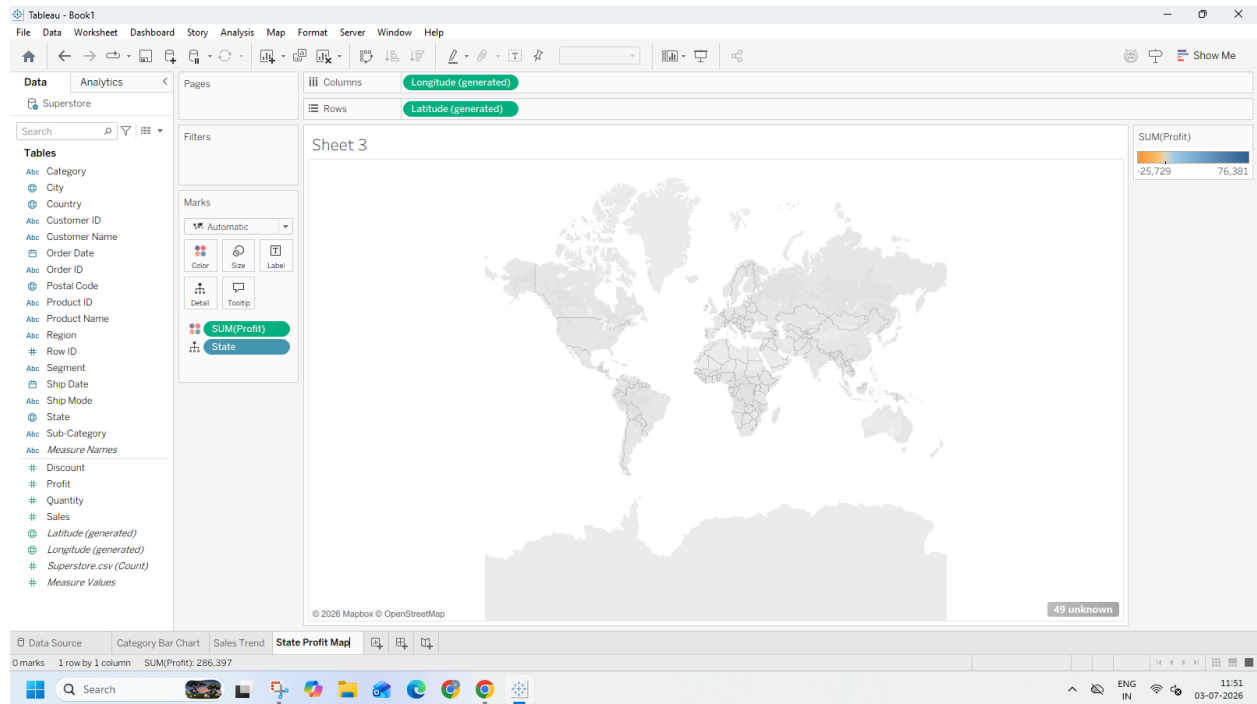
1. Dragging Order Date to Column and Sales into Row, then clicking the “+” sign on the Order Date in the Column Section to Divide into Quarters.



**f) Create a geographic map visualization:**

1. Double clicking the states to add the longitude and latitude into Column and Rows. And Dragging Profits into Color Tile in Marks card. And Naming the chart 'State Profit Map'.

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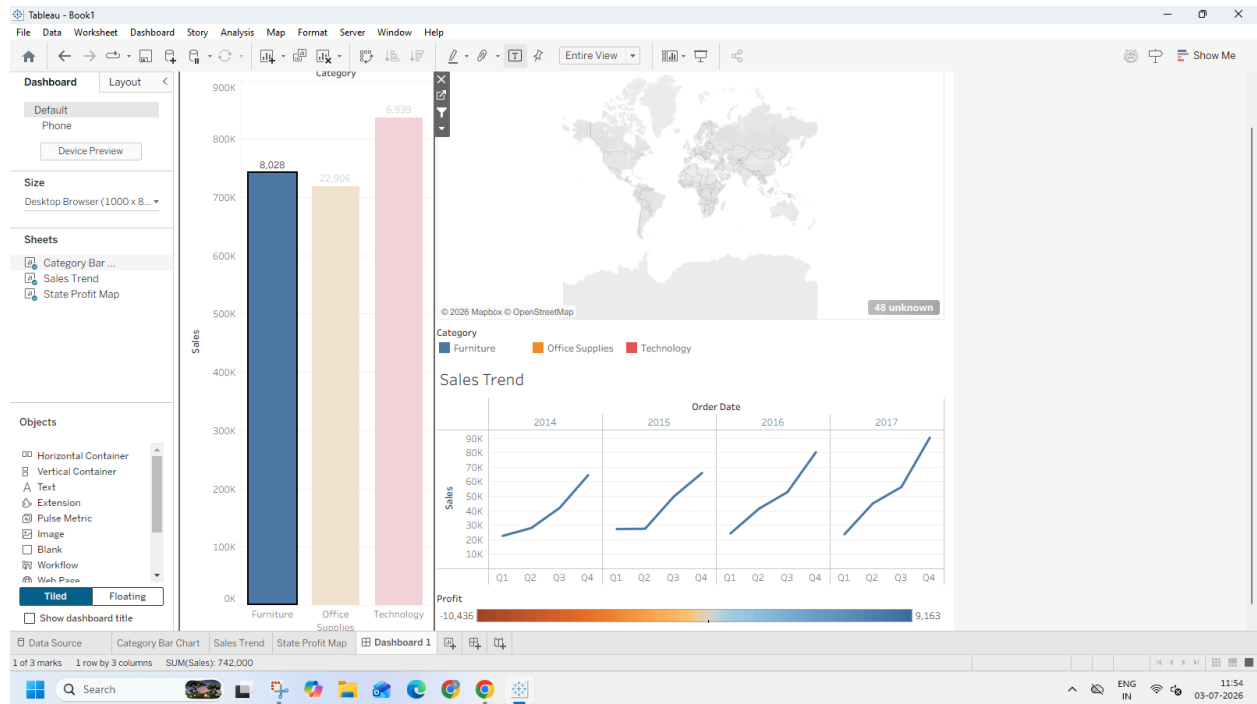
**g) Combine multiple charts into a simple interactive dashboard:**

1. Clicking bottom navigation tab bar and click the New Dashboard icon (it looks like a small 2x2 grid window pane with a plus symbol).
2. Dragging your Category Bar Chart sheet onto the empty dashboard canvas area.
3. Dragging your Sales Trend sheet and drop it right along the bottom edge of the bar chart workspace.
4. Dragging your State Profit Map and drop it alongside the right side of the layout.



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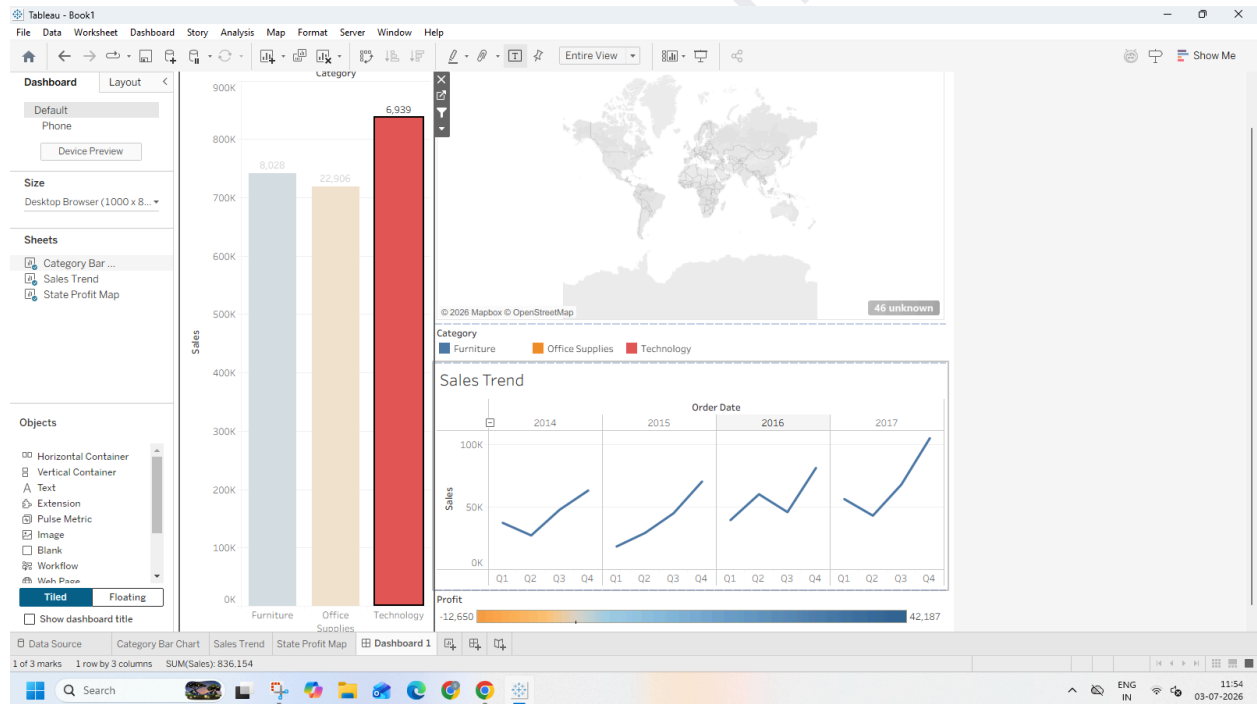
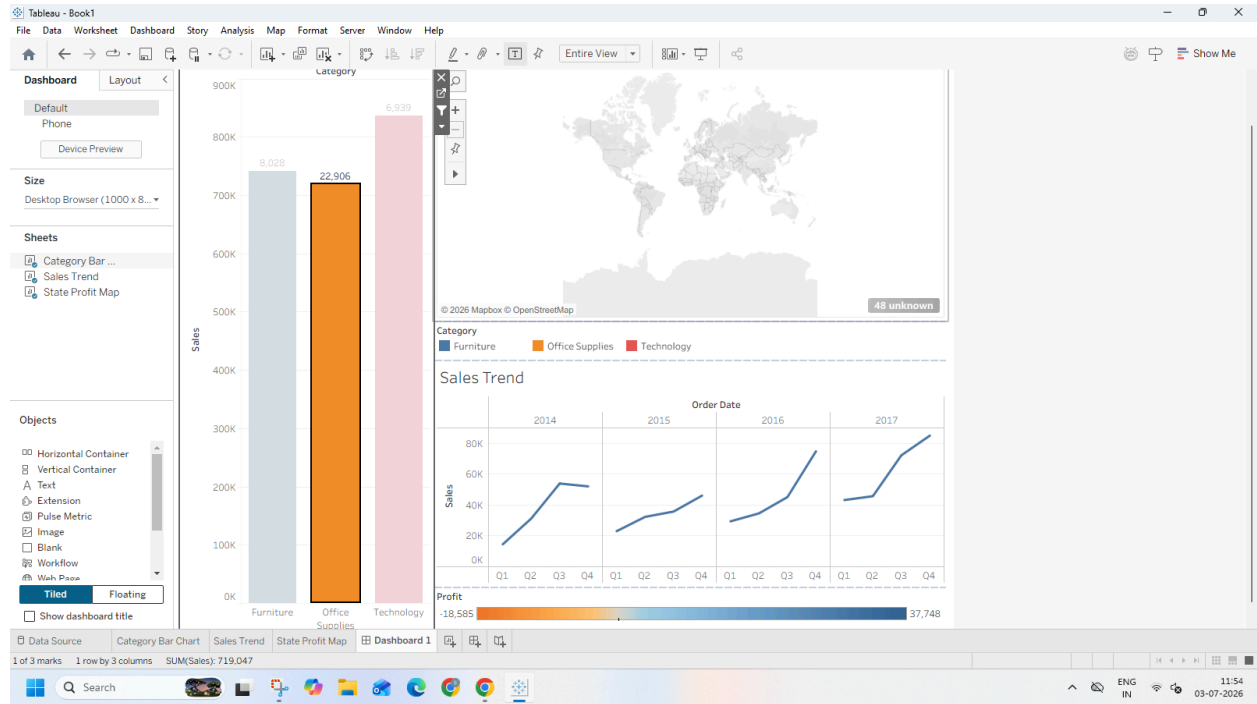
### Turning on the Magic: Interactivity

1. Clicking once inside Category Bar Chart zone on the dashboard to select it.
2. Clicking the tiny Funnel Icon (Use as Filter).

Now test out your creation! Click on the Technology bar in your bar chart. The map and the time-series line chart will dynamically isolate and update to display data exclusively for your Technology sales. Click the bar a second time to reset the filter.

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